



EXPLORATORY DATA ANALYSIS PROJECT

COSMETIC CHEMICALS

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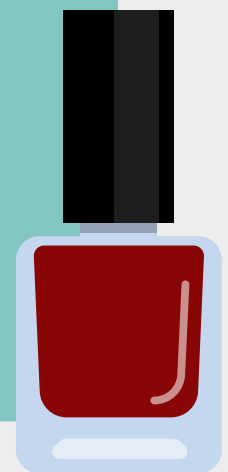
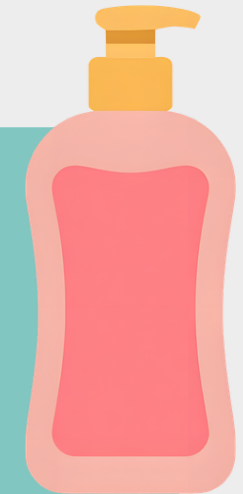
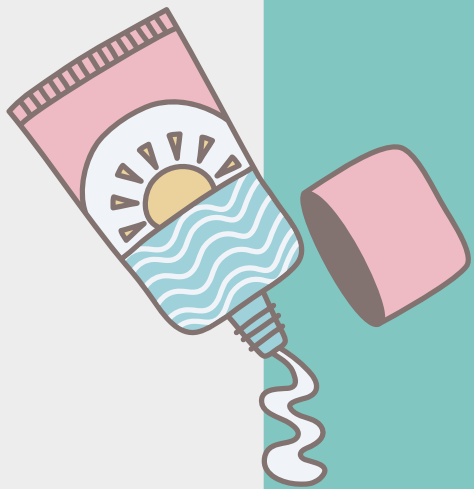
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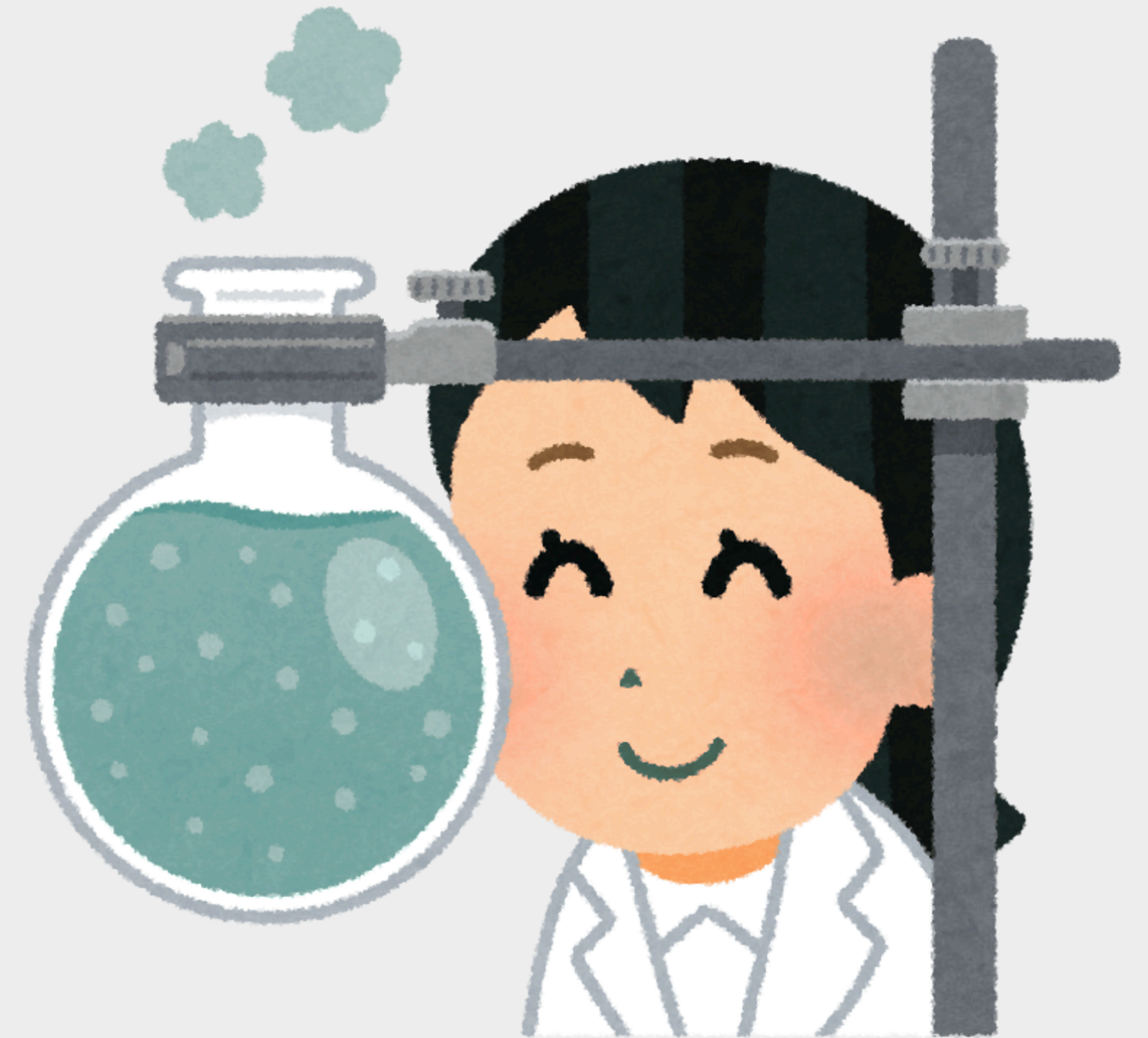
INTRODUCTION

The data is from the California Safe Cosmetics Program (CSCP) in the California Department of Public Health. The primary purpose of the CSCP is to collect information on hazardous and potentially hazardous ingredients in cosmetic products sold in California and to make this information available to the public. Many different chemicals are used in making cosmetics. Consumer and worker advocacy groups are concerned because some cosmetic products contain chemicals known or suspected to cause cancer, birth defects, or damage to the reproductive system. Those who work with cosmetics — including barbers, hair stylists, and skin care, body care and nail salon workers — may be more vulnerable to the adverse health effects posed by these products.

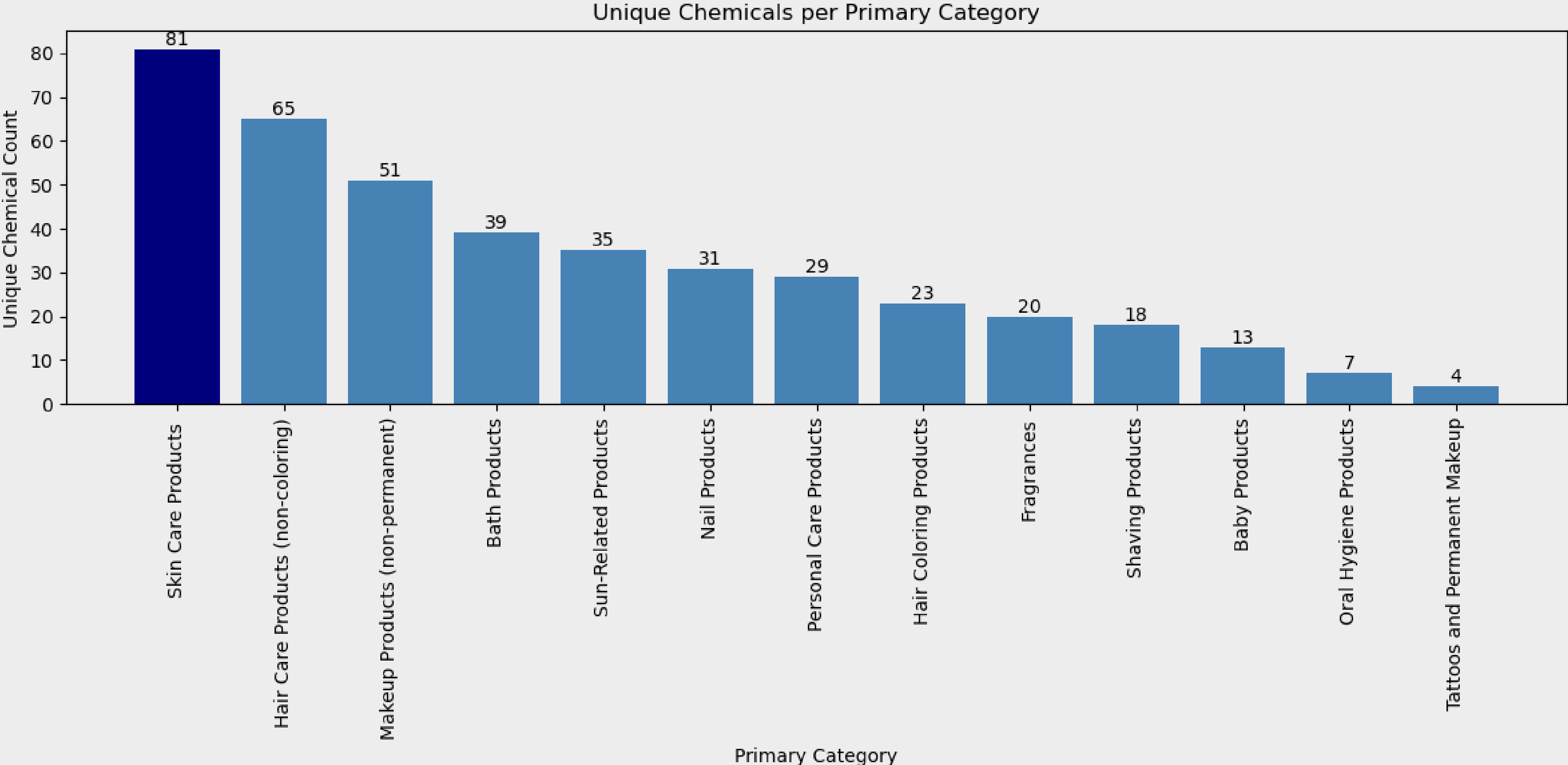


PROBLEM STATEMENT

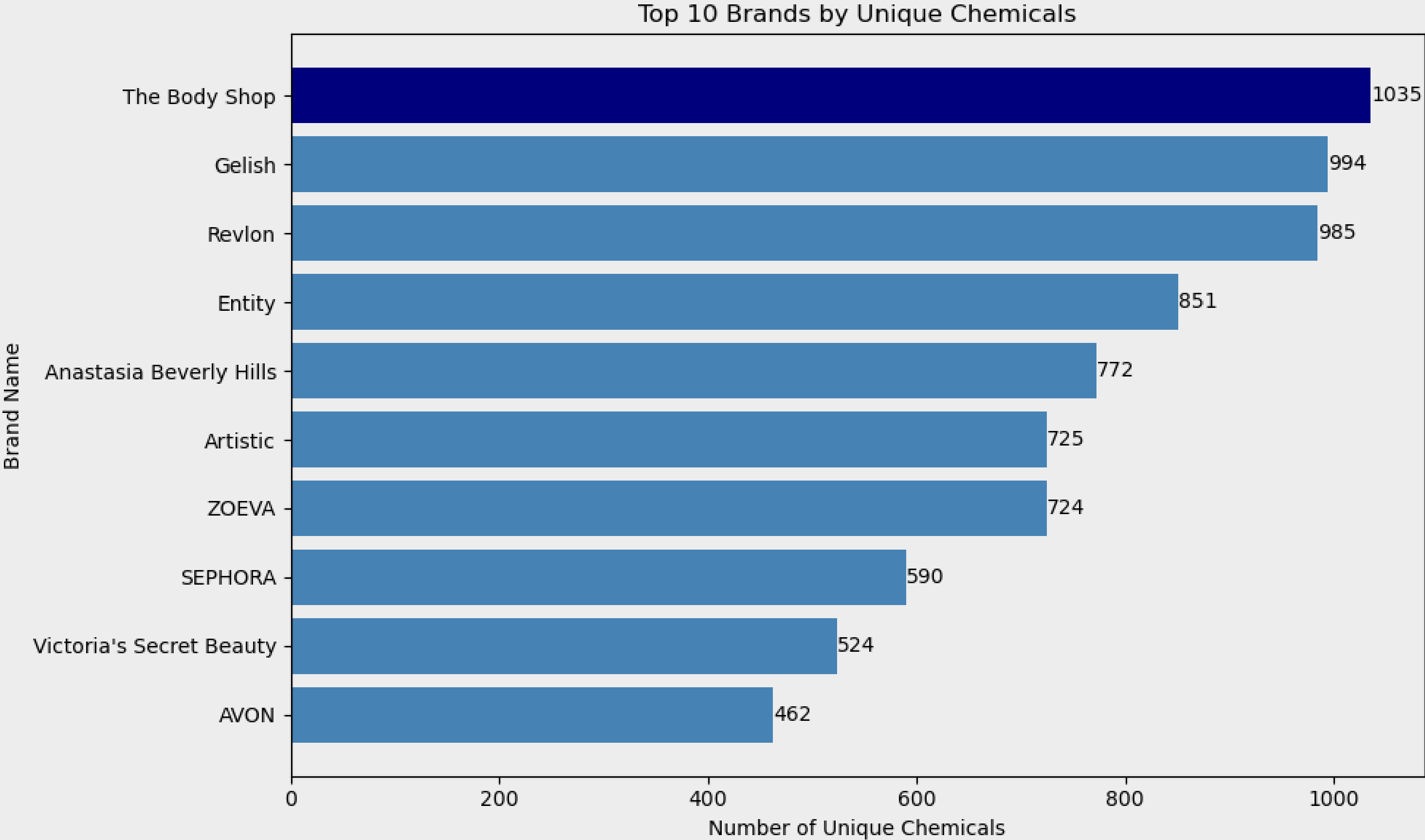
This study analyzes the reporting speed of hazardous chemicals in cosmetic products and their removal or reformulation under the California Safe Cosmetics Act. It also explores chemical diversity across product categories, identifies the most frequently reported chemicals, determines which Companies and brands use the widest variety of chemicals, and evaluates trends in reported, removed, and discontinued chemicals over time.



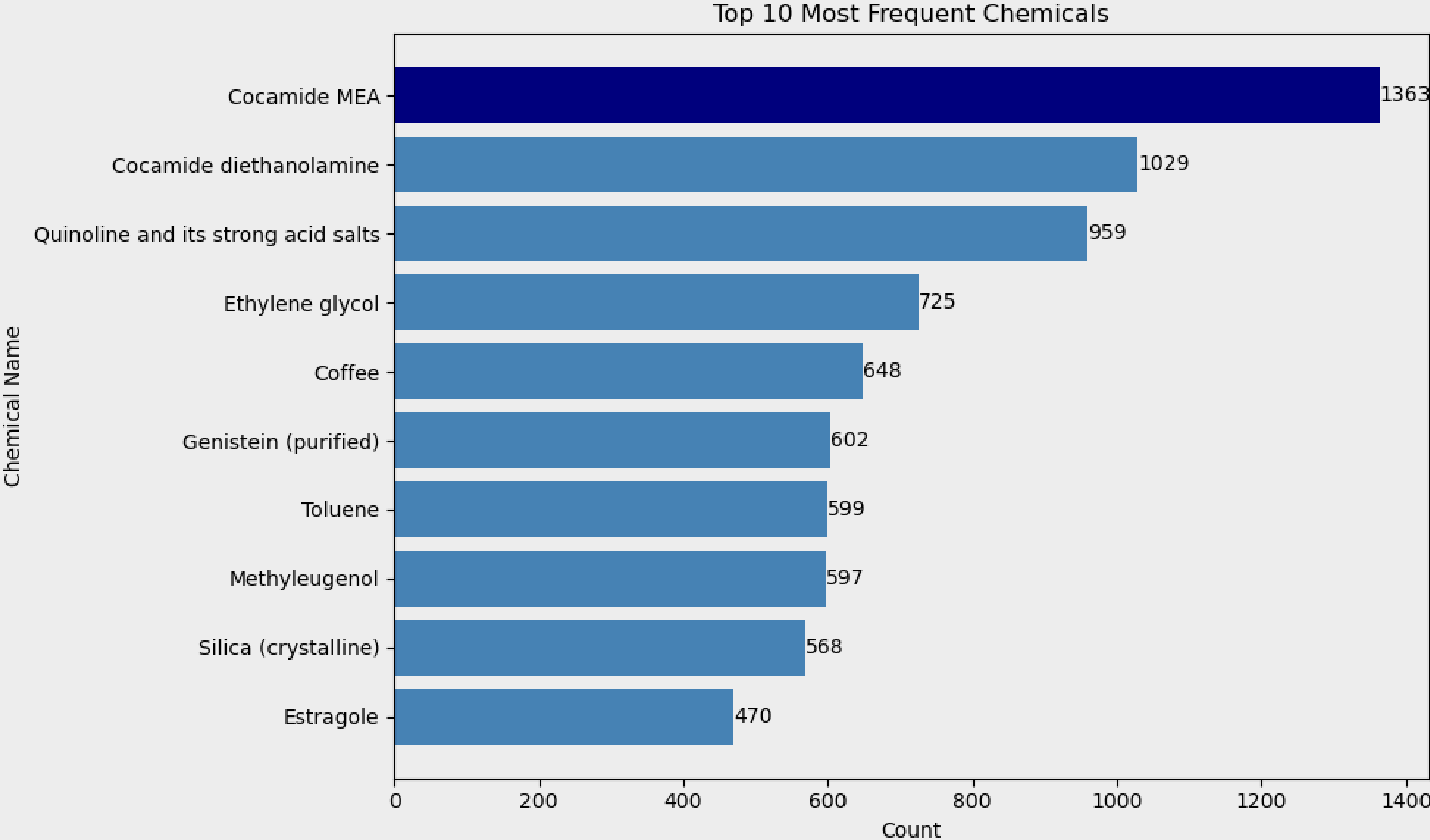
Which primary product categories contain the highest diversity of chemicals?



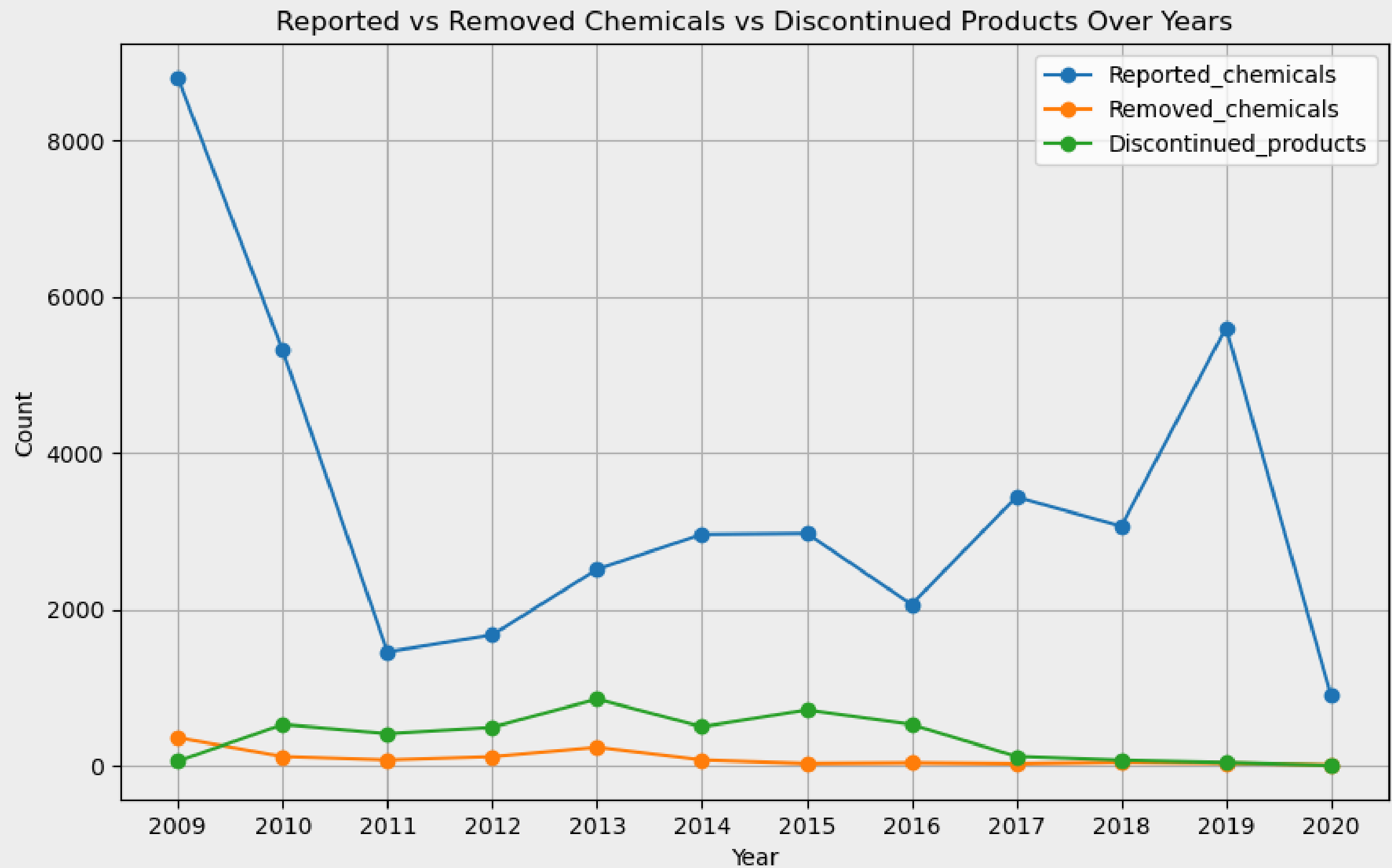
Which companies brands use the widest variety of chemicals?



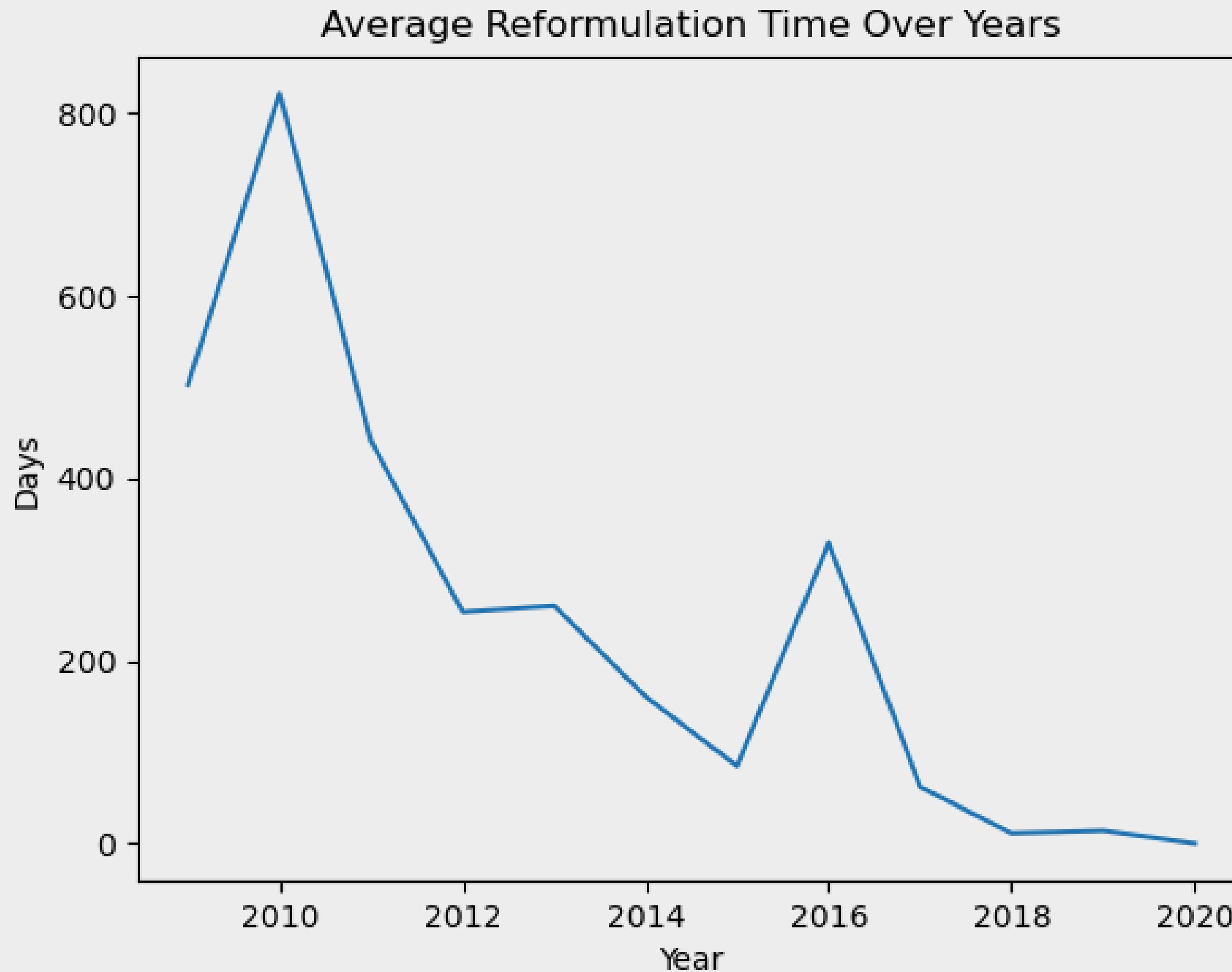
Which chemicals appear most frequently across cosmetic products?



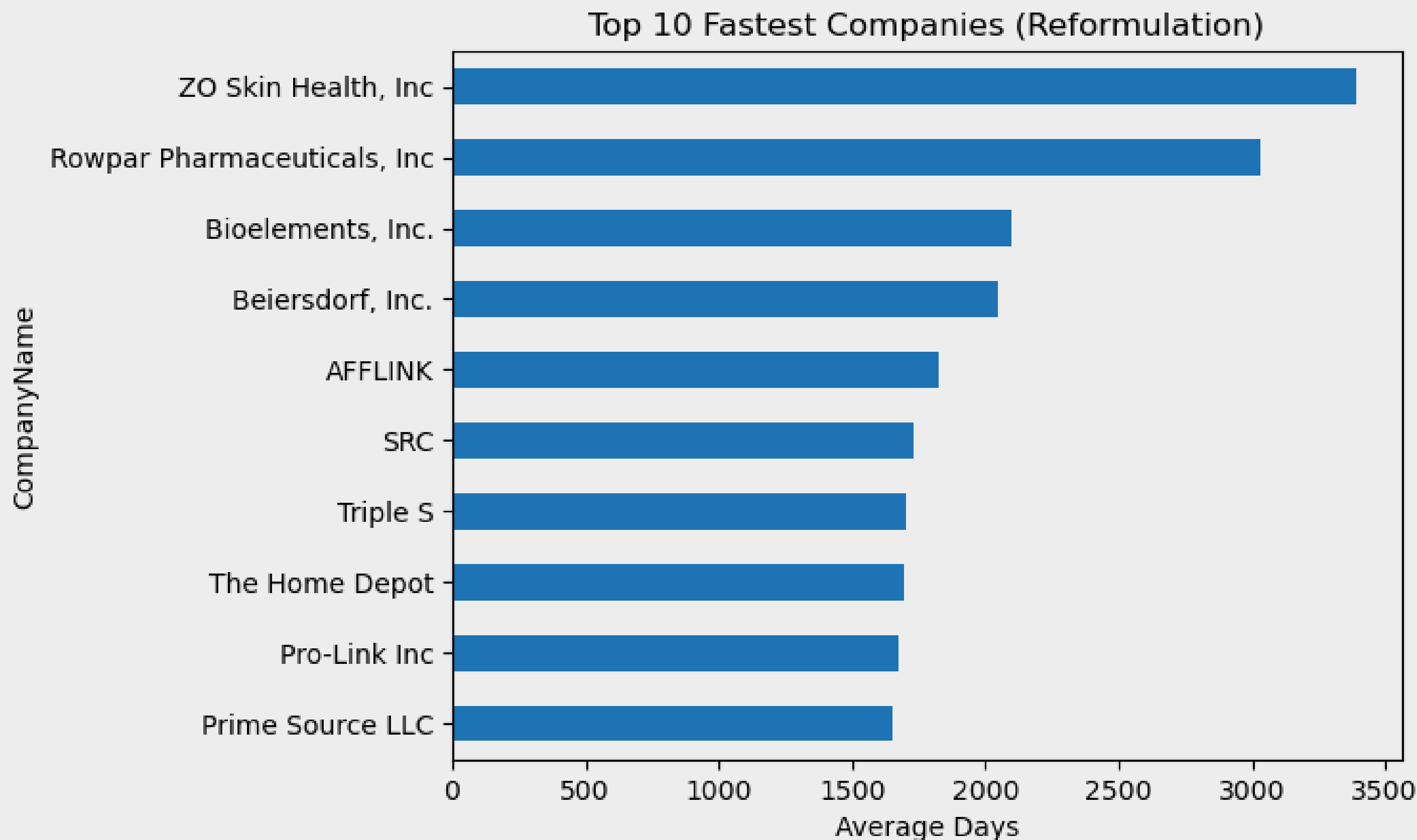
How have the number of reported chemicals, removed chemicals, and discontinued products changed over the years?



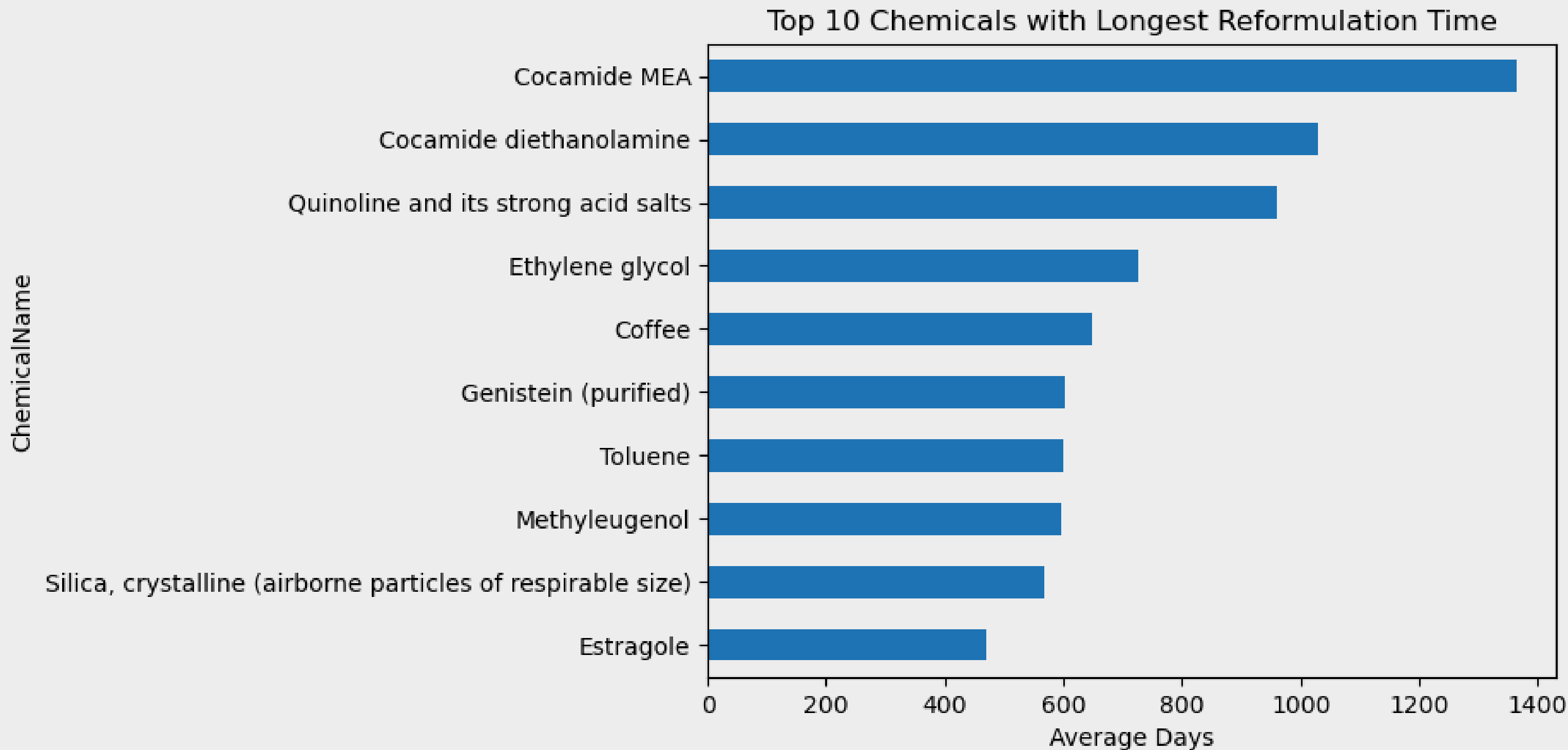
What is the Average Reformulation Time Over Years?



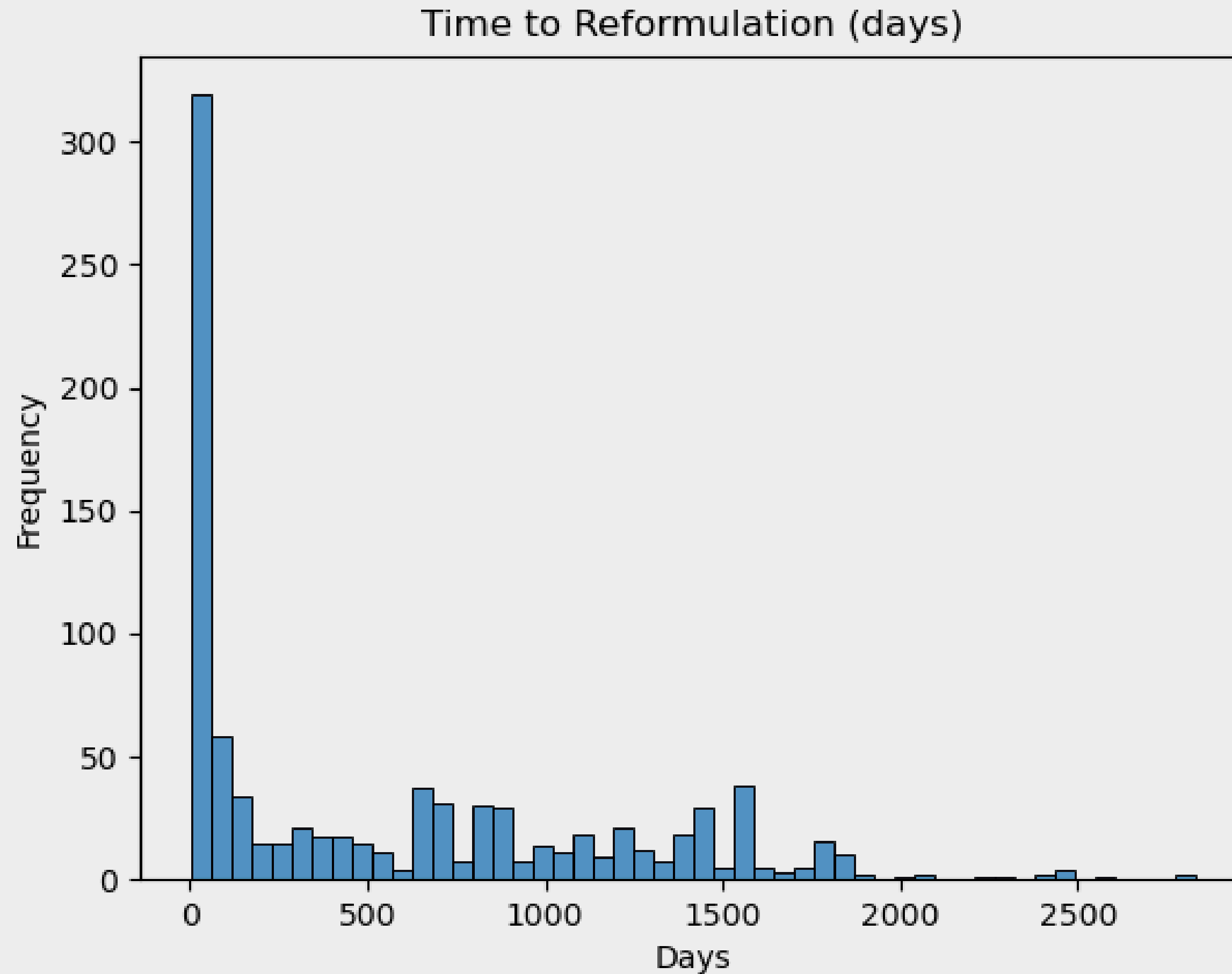
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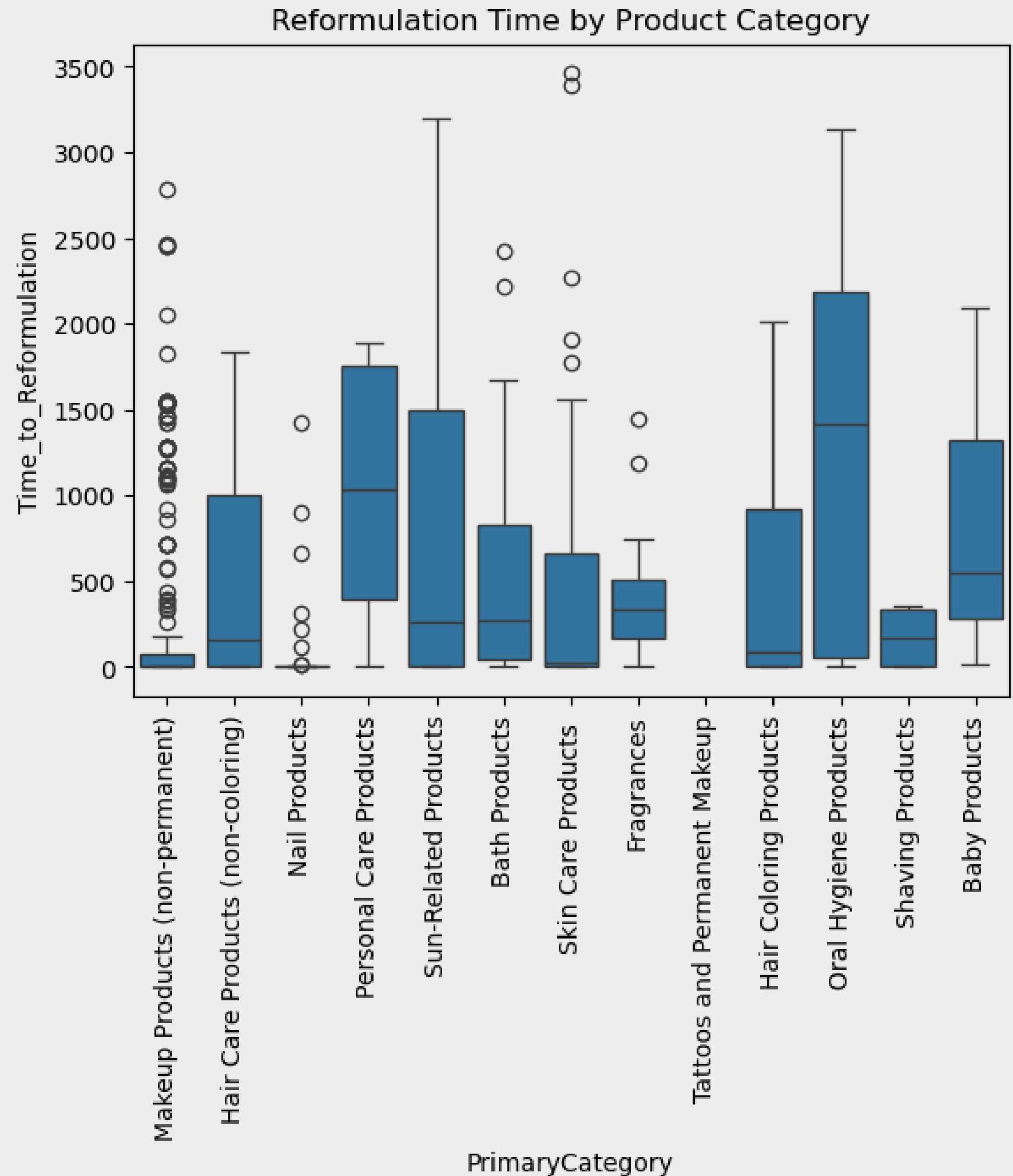
Which companies take the longest time to remove chemicals from their products (reformulation)?



Distribution of the time to remove chemicals from products (reformulation)



Reformulation Time by Product Category



KEY INSIGHTS

01

Skin care products contain the highest number of reported chemicals.

02

The most frequently reported chemicals are titanium dioxide and silica.

03

Revlon and The Body Shop have the highest number of unique chemicals.

04

Most chemicals are removed from products relatively quickly after being reported.

05

Nail products have the shortest reformulation time, while oral hygiene products have the longest.

06

Cocamide shows the longest reformulation time, suggesting it may be underregulated or overlooked.

07

ZO Skin Health, Inc. has the highest average reformulation time among companies.

08

From 2009 to 2020, the average reformulation time has decreased

RECOMMENDATIONS

01

Cocamide, which exhibit long reformulation times, should be prioritized for regulatory review and monitoring, as prolonged presence may indicate delayed industry response or challenges in substitution.

02

Skin care products should be subject to stricter formulation guidelines and transparency requirements, as they contain the highest number of reported chemicals and may pose greater cumulative exposure risks.

03

While many chemicals are removed quickly, further investigation is needed to determine whether this reflects proactive safety measures or delayed initial reporting.

04

The observed decrease in reformulation time suggests improved industry responsiveness; however, continued monitoring is necessary to ensure consistent compliance across all companies and chemical types.



**THANK YOU
VERY MUCH!**